

Consecutive 4-Cycle Graphs $G_4(k, n)$

Area lower bounds, residue obstructions, and empirical correction terms

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cc4-zk-n-graph-slides.tex

Problem setup

Vertices

$G_4(k, n)$ has vertices equal to binary words of length n with exactly k zeros and $n - k$ ones.

Canonical vertex

$$A = 0^k 1^{n-k}.$$

Edges

Edges are generated by cyclic rotations of consecutive length-4 blocks:

$$x_i x_{i+1} x_{i+2} x_{i+3} \mapsto x_{i+1} x_{i+2} x_{i+3} x_i,$$

together with the inverse rotations.

The basic statistics

Let B have zero positions

$$1 \leq z_1 < z_2 < \cdots < z_k \leq n.$$

Displacement vector

$$a(B) = (a_1, \dots, a_k), \quad a_i = z_i - i.$$

Inversion / area statistic

$$I(B) = \sum_{i=1}^k a_i.$$

Residue word for 4-cycles

$$r_i \equiv a_i \pmod{3}, \quad r(B) = r_1 r_2 \cdots r_k.$$

Area lower bound and correction term

A single consecutive 4-cycle can increase $I(B)$ by at most 3.

Universal lower bound

$$d_{G_4}(A, B) \geq \left\lceil \frac{I(B)}{3} \right\rceil.$$

This motivates the correction term

$$h_4(B) = d_{G_4}(A, B) - \left\lceil \frac{I(B)}{3} \right\rceil$$

so that

$$d_{G_4}(A, B) = \left\lceil \frac{I(B)}{3} \right\rceil + h_4(B).$$

The goal is to characterize when $h_4(B) = 0, 1, 2, \dots$

Why G_4 is harder than G_3

For $G_3(k, n)$, the distance from A has a clean formula

$$d_{G_3}(A, B) = \frac{I(B) + \Delta(B)}{2}.$$

For $G_4(k, n)$, the area bound is not always sharp.

Small obstruction

$$A = 00111, \quad B = 10101.$$

The zero positions of B are $(2, 4)$, hence

$$I(B) = (2 - 1) + (4 - 2) = 3, \quad \lceil I(B)/3 \rceil = 1.$$

But B is not reachable in one move:

$$00111 \rightarrow 01011 \rightarrow 10101, \quad d_{G_4}(A, B) = 2.$$

Thus $h_4(B) = 1$.

The obstruction persists under padding

The obstruction is local, so it survives in larger graphs.

For any $m, p \geq 0$, define

$$A' = 0^{m+2}1^{p+3}, \quad B' = 0^m101011^p.$$

Then

$$I(B') = 3, \quad \lceil I(B')/3 \rceil = 1, \quad d_{G_4}(A', B') = 2.$$

Consequence

There is no large- k or balanced-regime simplification where

$$d_{G_4}(A, B) = \lceil I(B)/3 \rceil$$

becomes universally exact.

Closed cases: $k = 1, 2, 3$

$k = 1$

If $a_1 = 3q + r$, $r \in \{0, 1, 2\}$, then

$$d_{G_4}(A, B) = q + r, \quad h_4(B) = 1 \iff a_1 \equiv 2 \pmod{3}.$$

$k = 2$

$$d_{G_4}(A, B) = \left\lceil \frac{a_1 + a_2 + (a_1 \bmod 3)}{3} \right\rceil.$$

Equivalently,

$$h_4(B) = 1 \iff (a_1, a_2) \bmod 3 \in \{(1, 2), (2, 0), (2, 1)\}.$$

$k = 3$

For $n - k \geq 4$,

$$h_4(B) = 1 \iff \{r_1, r_2, r_3\} = \{0, 1, 2\}.$$

Closed case: $k = 4$

Let $r(B) = r_1 r_2 r_3 r_4$ be the residue word, where $r_i = a_i \bmod 3$.

For $k = 4$ and $n - k \geq 3$,

$$d_{G_4}(A, B) = \left\lceil \frac{I(B)}{3} \right\rceil + \mathbf{1}_{r(B) \in \mathcal{E}_4}.$$

Here $\mathcal{E}_4$ is the set of 30 exceptional residue words.

Interpretation

For $k = 4$, the correction is still a 0/1 phenomenon, but the obstruction is no longer captured by one simple parity condition. It is controlled by residue words modulo 3.

Sharp families with $h_4(B) = 0$

Some broad families attain the lower bound exactly.

All displacements divisible by 3

If

$$a_i \equiv 0 \pmod{3} \quad \text{for all } i,$$

then

$$d_{G_4}(A, B) = \frac{I(B)}{3}, \quad h_4(B) = 0.$$

Suffix block shifted by t

If

$$a = (0, \dots, 0, \underbrace{t, \dots, t}_{p \text{ times}}),$$

then

$$d_{G_4}(A, B) = \left\lceil \frac{pt}{3} \right\rceil, \quad h_4(B) = 0.$$

First examples with $h_4(B) = 2$

For $k \geq 5$, the correction can exceed 1.

A clean $k = 5$ example

Take

$$a = (2, 4, 6, 8, 10), \quad r(B) = 21021.$$

Then

$$I(B) = 30, \quad \lceil I(B)/3 \rceil = 10,$$

but the exact distance is

$$d_{G_4}(A, B) = 12.$$

Therefore

$$\boxed{h_4(B) = 2.}$$

Equivalently, for $B = (110)^5$ with $k = 5, n = 15$, the same correction occurs.

Empirical distribution of $h_4(B)$

Examples over the component C_A of the canonical vertex:

Graph	$ C_A $	D_A	$h = 0$	$h = 1$	$h = 2$
$G_4(2, 5)$	10	3	8	2	0
$G_4(4, 8)$	70	6	57	13	0
$G_4(6, 12)$	924	12	681	243	0
$G_4(8, 16)$	12,870	22	7,017	5,832	21
$G_4(10, 20)$	184,756	34	96,265	86,931	1,560

Pattern

Most vertices have $h_4(B) = 0$ or $h_4(B) = 1$. Higher corrections appear only in larger, more balanced examples.

Large balanced example: $G_4(13, 27)$

For $(k, n) = (13, 27)$, the computed distribution is:

$h_4(B)$	count	percentage
0	7,568,546	37.73%
1	11,955,665	59.60%
2	531,946	2.65%
3	2,142	0.0107%

Takeaway

Even in this large balanced case, more than 97% of vertices have

$$h_4(B) \in \{0, 1\}.$$

Diameter observations

Define

$$D_A(G_4(k, n)) = \max_B d_{G_4}(A, B).$$

Computed examples:

(k, n)	$m = n - k$	$ V = \binom{n}{k}$	$D_A(G_4(k, n))$
$(4, 10)$	6	210	8
$(9, 20)$	11	167,960	33
$(12, 25)$	13	5,200,300	52

In all three cases,

$$D_A(G_4(k, n)) = \left\lceil \frac{k(n-k)}{3} \right\rceil.$$

The farthest layer has h_4 values

$$\{0, 0, 0, 0, 1\}.$$

Normalized correction

For $B \neq A$, define

$$h'_4(B) = \frac{h_4(B)}{d_{G_4}(A, B)} = \frac{d_{G_4}(A, B) - \lceil I(B)/3 \rceil}{d_{G_4}(A, B)}.$$

Interpretation

$h'_4(B)$ measures the fraction of the true distance contributed by the correction beyond the area lower bound.

Empirically, for balanced examples $G_4(k, 2k)$:

- $h_4(B)$ is common, especially $h_4(B) = 1$.
- $h'_4(B)$ is usually small once distances become moderately large.
- Large relative corrections mostly come from short-distance exceptional vertices.

Small- k quasi-polynomial patterns

For fixed k , the counts

$$H_j(k, n) = \#\{B \neq A : h_4(B) = j\}$$

show period-3 structure in n .

$k = 1$

$$H_1(1, n) = \left\lfloor \frac{n}{3} \right\rfloor, \quad H_0(1, n) = n - 1 - H_1(1, n).$$

$k = 2$

$$H_1(2, n) = \begin{cases} \frac{n(n-3)}{6}, & n \equiv 0 \pmod{3}, \\ \frac{(n-1)(n-2)}{6}, & n \equiv 1, 2 \pmod{3}. \end{cases}$$

Higher correction families

Some higher obstructions also show period-3 behavior.

First visible $h_4 = 2$ family

For $k = 5$, the data is consistent with

$$H_2(5, n) = \binom{\lfloor n/3 \rfloor}{5}.$$

First visible $h_4 = 3$ family

For $k = 8$, the data is consistent with

$$H_3(8, n) = \binom{\lfloor n/3 \rfloor}{8}.$$

This suggests that some higher obstruction families are counted by choosing special blocks among roughly $\lfloor n/3 \rfloor$ available positions.

Working conjecture

Conjecture

For fixed k and fixed correction level j , the count

$$H_j(k, n)$$

is eventually a quasi-polynomial in n , likely with period 3.

Equivalently, after splitting by

$$n \bmod 3,$$

each subsequence should eventually agree with a polynomial in n .

Why period 3 is natural

A consecutive 4-cycle can change the inversion/area statistic by at most 3, and the residue word records displacements modulo 3.

- $G_4(k, n)$ is generated by consecutive length-4 rotations on binary words.
- The area statistic gives the universal lower bound

$$d_{G_4}(A, B) \geq \lceil I(B)/3 \rceil .$$

- The correction

$$h_4(B) = d_{G_4}(A, B) - \lceil I(B)/3 \rceil$$

is controlled strongly by residues $a_i \bmod 3$.

- For most computed vertices, $h_4(B) = 0$ or 1 .
- Higher corrections exist, but appear sparse and structured.
- The main empirical direction is a period-3 quasi-polynomial theory for $H_j(k, n)$.